

# Statistical Machine Learning

## $p$ -values and Multiple Testing

Horacio Gómez-Acevedo  
Department of Biomedical Informatics  
University of Arkansas for Medical Sciences

February 26, 2025



# What is a Hypothesis test?

- A *statistical hypothesis*  $H$  is a conjecture about the probability distribution of a population.
- A hypothesis  $H$  is said to be *simple* if the distribution of the population is completely specified by  $H$ . If not, then  $H$  is called a *composite* hypothesis.
- $H_0$  is called the *null hypothesis* and  $H_1$  the *alternative hypothesis*.
- We say that we commit a *type I error* if we decide to accept  $H_1$ , whereas in reality  $H_0$  is true.
- The probability of committing a type I error will be denoted by  $\alpha$ .
- Acceptance of  $H_0$  whereas  $H_1$  is true is called a *type II error*.
- The probability of committing a type II error will be denoted by  $\beta$ .

# Error types

| Decision                     | True State of Nature |                     |
|------------------------------|----------------------|---------------------|
|                              | $H_0$ is TRUE        | $H_1$ is TRUE       |
| Reject $H_0$<br>Accept $H_1$ | TYPE I<br>Error      | Correct<br>Decision |
| Accept $H_0$<br>Reject $H_1$ | Correct<br>Decision  | TYPE II<br>Error    |

## Example 1

A factory has packets of coffee with and adjusted weight of 500 grams. We assume that the weight of the packages is  $N(500, 50)$ -distributed. Coffee packages are stored in containers (of unknown number). An error in a scale, some of the containers are filled up with packages of weight  $N(490, 50)$ -distributed. To determine containers that have incorrectly weight the packages, we draw a sample of two packages from the given container. Based on the outcome of a 2-vector  $(X_1, X_2)$  we will make a conjecture about the packages in that container. Namely

$H_0$ : the population is  $N(500, 50)$  – distributed

$H_1$ : the population is  $N(490, 50)$  – distributed

Let's suppose our decision as

*If both  $X_1$  and  $X_2 \leq 496$  then we accept  $H_1$ , otherwise we accept  $H_0$ .*

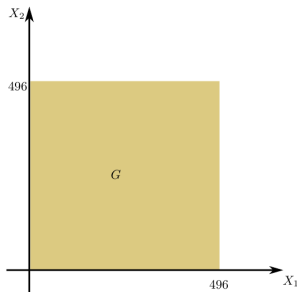
## Example 1 (cont)

Let's define our *critical region*  $G$  as

$$G = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1, x_2 \leq 496\}$$

Then, we can formulate our decision as

$$= \begin{cases} \text{if } (X_1, X_2) \in G & \text{then we choose } H_1 \text{ as our conjecture.} \\ \text{if } (X_1, X_2) \notin G & \text{then we choose } H_0 \text{ as our conjecture.} \end{cases}$$



## Calculating $\alpha$ and $\beta$

$$\begin{aligned}\alpha &= P(\text{acceptance of } H_1 | H_0 \text{ is true}) \\ &= P((X_1, X_2) \in G | \mu = 500, \sigma^2 = 50) \\ &= P(X_1 \leq 496 | \mu = 500, \sigma^2 = 50) \cdot P(X_2 \leq 496 | \mu = 500, \sigma^2 = 50) \\ &= 0.081\end{aligned}$$

$$\begin{aligned}\beta &= P(\text{acceptance of } H_0 | H_1 \text{ is true}) \\ &= P((X_1, X_2) \notin G | \mu = 490, \sigma^2 = 50) \\ &= 1 - P((X_1, X_2) \in G | \mu = 490, \sigma^2 = 50) \\ &= 1 - P(X_1 \leq 496 | \mu = 490, \sigma^2 = 50) \cdot P(X_2 \leq 496 | \mu = 490, \sigma^2 = 50) \\ &= 0.356\end{aligned}$$

# Hypothesis test

A *Hypothesis test* is a collection

$$(X_1, \dots, X_n; H_0; H_1 : G)$$

where  $X_1, \dots, X_n$  is a sample,  $H_0$  and  $H_1$  hypotheses concerning the probability distribution of the population and  $G \subset \mathbb{R}^n$  a Borel set (meaning a collection of open sets)

If  $H_0$  is a simple statistical hypothesis. The *level of significance* of the hypothesis test  $(X_1, \dots, X_n; H_0; H_1; G)$  is understood to be the number

$$\alpha = P_{X_1, \dots, X_n}^{H_0}(G)$$

Thus, we say that  $\alpha$  represents the probability of committing a type I error.

# The power function

With our previous setup the  $\beta$  could not be used for composite hypothesis. Thus, we have to define something more general.

- Let  $f(\cdot, \theta)_{\theta \in \Theta}$  be a family of probability densities
- Let's assume that the population  $X_1, \dots, X_n$  has a probability density  $f(\cdot, \theta)$  where  $\theta \in \Theta$ .
- Let's assume that  $H_0$  and  $H_1$  are statements of the type

$$H_0: \theta \in \Theta_0 \quad \text{and} \quad H_1: \theta \in \Theta_1$$

where  $\Theta_0 \cup \Theta_1 = \Theta$  and  $\Theta_0 \cap \Theta_1 = \emptyset$ .

For a fixed  $\theta \in \Theta$ , the probability distribution of the population is completely specified. Then, we define for every  $\theta \in \Theta_1$

$$\beta(\theta) = P_{X_1, \dots, X_n}^{\theta}(G^c)$$

The expression  $1 - \beta(\theta)$  is called the *power function* for  $\theta \in \Theta_1$ .



## Example 2

From our Example 1, given is a  $N(\mu, 50)$ -distributed population, where  $\mu \leq 500$ . The family of probability densities  $f(\cdot, \mu)$  where  $\mu \leq 500$  and

$$f(x, \mu) = \frac{1}{\sqrt{100\pi}} \exp\left(\frac{-(x - \mu)^2}{100}\right)$$

The parameter space is defined by  $\Theta = (-\infty, 500]$ .

If we draw a sample  $X_1, X_2$  of size 2 from this population. We can define  $\Theta_0 = \{500\}$  and  $\Theta_1 = (-\infty, 500)$ . This corresponds to the following hypotheses

$$H_0: \mu = 500 \quad \text{against} \quad H_1: \mu < 500$$

If we choose the following critical region

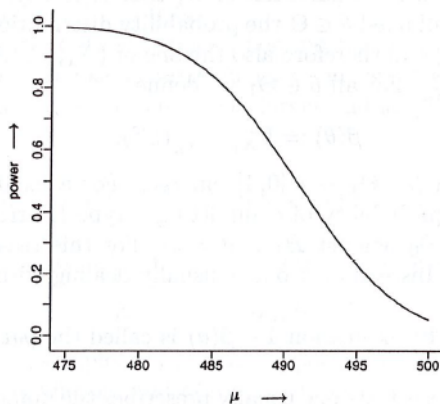
$$G = \{(x_1, x_2) \in \mathbb{R}^2 : x_1, x_2 \leq 494.63\}$$

## Example 2(cont)

Then, our 5-tuple  $(X_1, X_2; H_0; H_1; G)$  constitutes a hypothesis test.  
The size  $\alpha$  of the critical region  $G$  is  $\alpha = 0.05$ , and the power function

$$\begin{aligned} 1 - \beta(\tilde{\mu}) &= 1 - P_{X_1, X_2}^{\mu}(G^c) = P_{X_1, X_2}^{\mu}(G) = P((X_1, X_2) \in G) \\ &= P(X_1 \leq 494.63 \text{ and } X_2 \leq 494.63 | \mu = \tilde{\mu}, \sigma^2 = 50) \\ &= P(X_1 \leq 494.63 | \mu = \tilde{\mu}, \sigma^2 = 50) \\ &\quad \cdot P(X_2 \leq 494.63 | \mu = \tilde{\mu}, \sigma^2 = 50) \end{aligned}$$

# Power function example



## Normally Distributed Case

Suppose we are dealing with a  $N(\mu, \sigma^2)$ -distributed population where both  $\mu$  and  $\sigma$  are unknown. If we wish to test the hypothesis

$$H_0: \mu = \mu_0 \text{ against } H_1: \mu \neq \mu_0$$

Critical regions based on the likelihood ratio are of type

$$G = \{(x_1, \dots, x_n) \in \mathbb{R}^n : \left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| \geq c\}$$

where  $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

The outcome of the variable

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

is decisive in the choice to reject  $H_0$  or not. This is called the *Test statistic* for this hypothesis test. Under the  $H_0$  the test statistics is *t*-distributed with  $n - 1$  degrees of freedom.

## Normally distributed case

The decision becomes

if  $|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}| \geq c$ , then assume  $H_1$

if  $|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}| < c$ , then assume  $H_0$

And we have the following equivalence

$$P(|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}| \geq u) \leq \alpha \iff u \geq c$$

If the outcome  $u$  of  $|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}|$  satisfies

$$P(|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}| \geq |u|) \leq \alpha$$

then, we accept  $H_1$ .

## $p$ -values

The expression

$$P(|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}| \geq |u|)$$

is called the  $p$ -value associated with the outcome  $u$  of the test statistic.  
Therefore, we normally represent the decision rule

if  $P\text{-value} \leq \alpha$ , then assume  $H_1$

if  $P\text{-value} > \alpha$ , then assume  $H_0$

We were familiar from basic statistics courses about the normally distributed instances. But from the previous discussion we can see that the same ideas can expand to accommodate

- different critical regions,
- different test statistics

# Cherry Picking Problem

Let's suppose we use the same data set and apply different test statistics at the  $\alpha$  level of significance:

- $W_\alpha$  a Wilcoxon test,
- $P_\alpha$  a permutation test, and
- $T_\alpha$  a  $t$ -test.

In this scenario, it may be possible that  $W_\alpha$  may be true when  $P_\alpha$  and  $T_\alpha$  are not.

$$P(W_\alpha \text{ or } T_\alpha \text{ or } P_\alpha | H) \geq P(W_\alpha | H) = \alpha$$

we will have an **inflated** type I error by picking and choosing after the fact which test to report.



# Cherry Picking Problem

Similarly, if our intent was to conceal a side effect by reporting the results were not significant, we will inflate the Type II error and **deflate** the power  $\beta$  of our test by an after-the-fact choice as

$$\beta = P(\text{not}(W_\alpha \text{ and } P_\alpha \text{ and } T_\alpha) | K) \leq (P(\text{not } W_\alpha | K))$$

This quote from (Good and Harding, 2006) is important to remember.  
*We are not free to pick and choose among tests; any such conduct is unethical.* **Both the comparison and the test statistic must be specified in advance of examining the data.**

# Multiple Testing

Consider the problem of simultaneously testing  $m$  null hypotheses  $H_j$ ,  $j = 1, \dots, m$  and denote by  $R$  the number of rejected hypothesis. In the frequentist setting, the situation can be summarized as

|                             | No. not rejected | No. rejected |       |
|-----------------------------|------------------|--------------|-------|
| No. true null hypotheses    | $U$              | $V$          | $m_0$ |
| No. nontrue null hypotheses | $T$              | $S$          | $m_1$ |
|                             | $m - R$          | $R$          | $m$   |

## Multiple Testing (cont)

The  $m$  hypotheses are assumed to be known in advance, while the number  $m_0$  and  $m_1$  of true and false null hypothesis are unknown parameter,  $R$  is an observable random variable, and  $S$ ,  $T$ ,  $U$ , and  $V$  are unobservable random variables.

In general, we want to minimize the number  $V$  of false positives, or type I errors, and the number  $T$  of false negative or type II errors. The standard approach is to pre-specify an acceptable type I error rate  $\alpha$  that seek tests that minimize the type II error rate, within the class of tests with type I error rate  $\alpha$ .

# Test for multiple comparisons

In terms of these random variables, we can define the main rates used in the present context. When we are facing testing hypotheses of possibly thousands of significance tests, there are a number of alternatives dealing

- The per-comparison error rate (PCER). The expected value of the number of type I error over the number of hypotheses

$$PCER = \mathbb{E}(V)/m$$

- The family-wise error rate (FWER). The probability of at least one type I error

$$P(V \geq 1)$$

- The false discovery rate (FDR) is the expected proportion of type I error among rejected hypotheses

$$FDR = \mathbb{E}(V/R; R > 0) = \mathbb{E}(V/R | R > 0)P(R > 0)$$

## Adjusted $p$ -values

To account for multiple hypothesis testing, one may calculate the adjusted  $p$ -values. Given a test procedure, the adjusted  $p$ -value corresponding to the test of a single hypothesis  $H_j$  can be defined as the level of the entire test procedure at which  $H_j$  would just be rejected, given the values of all test statistics involved.

Control of the FWER at level  $\alpha$ , the Bonferroni procedure rejects any hypothesis  $H_j$  with  $p$ -value less than or equal to  $\alpha/m$ .

Note that we assume that the individual tests are independent of each other.

# Bonferroni correction

There are some issues with the Bonferroni's correction.

- In genomics, if you have 20,000 gene transcripts results for both controls and a certain disease. The Bonferroni correction a  $p$ -value would have to be less than  $0.05/20000 = 2.5 \times 10^{-6}$ . Your results might have a large number of false negatives from transcripts that might have been biologically important.
- Another issue is with the definition of a "family", so if you have a model with certain  $n_1$  dietary variables and  $n_2$  socioeconomic variables. The critical value should be  $0.05/n_1$  or  $0.05/(n_1 + n_2)$ ?

# FDR control

The Benjamini and Hocheberg procedure first puts all  $p$ -values in order from smallest to largest and ranks them. The critical value is defined as

$$(i/m)Q,$$

where  $i$  is the rank,  $m$  is the total number of tests, and  $Q$  is the **False Discovery Rate** that the user picks.

This adjustment leads to strong control of the FDR under the additional assumption of independence of the test statistics.

# Correct or not correct that is the question...

Whereas we aim use multiple comparison corrections to reduce the number of false positives, we also increase the number of false negatives! If false negatives are very costly (not only financially but scientifically), then you may decide not to correct for multiple comparison. For instance, if you have a small animal experiment using transcriptomics for the comparison of a case and control. If your conditions are very stringent for multiple comparison adjustments, you may not be able to find other possible interactions that may be occurring. Sometimes, you cannot even conduct a "pathway analysis" if you have very few hits.



# References

Materials and some of the pictures are from (Pestman, 1998), (MacDonald, 2014) and (Speed, 2000).

-  Good, Phillip I. and James W. Harding (2006). *Common Errors in Statistics. (and how to avoid them)*. John Wiley and Sons. ISBN: 978-0471794318.
-  MacDonald, John H. (2014). *Handbook for Biological Statistics*. Third Edition. Sparky House Publishing.
-  Pestman, W. R. (1998). *Mathematical Statistics. An Introduction*. De Gruyter. ISBN: 3-11.015356-4.
-  Speed, T., ed. (2000). *Statistical Analysis of Gene Expression Microarray Data*. CRC Press. ISBN: 978-1584883272.

I have used some of the graphs by hacking TiKz code from StakExchange, Inkscape for more aesthetic plots and other old tricks of T<sub>E</sub>X